Rare Common-Shock Detection from Binary Snapshots

Minimum cover order and optimal inspection

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Abstract

We compare two rates for a global shock in a system with independent subset shocks. Each replicate is inspected once, yielding the full binary status vector at a chosen time. We vary the global rate while compensating the dedicated rates, so the two candidates have identical one-dimensional margins. In fixed dimension, when the minimum cover uses $r \geq 3$ nonglobal labels and its leading reliability coefficient stays bounded away from zero, we derive the exact asymptotic squared Hellinger separation and characterize every first-order optimal inspection time. The asymptotics are uniform even when the difference of the two vanishing rates is arbitrarily small relative to their sum. At $r = 2$, the leading coefficient is one fourth of the maximum of the baseline Fisher curve over the compactified time axis. A fixed-dimensional example exhibits a strict gain at positive time. For two overlapping background shocks in growing dimension, we give two separation rates and an example in which their ratio diverges at matched global and background rates, total block count, and marginal calibration.

1 Introduction

Suppose that a replicated system is inspected once after a prescribed exposure. The status of every block is recorded, but the order and timing of individual events are unavailable. A global shock can produce the same all-complete snapshot as several smaller shocks occurring before inspection. Inspecting too early misses the global event. Inspecting too late lets other mechanisms produce the same observation. We ask how the organization of the background determines the best inspection time for distinguishing two candidate global rates.

Marshall and Olkin constructed multivariate lifetimes from independent subset shocks. At any fixed mission time the snapshot law belongs to the multivariate Bernoulli shock family studied by Boyles and Samaniego, with the local and dedicated mechanisms combined into one singleton activation probability per block. Their work includes arbitrary subset shocks, likelihood estimation, and simulations at rare-shock parameter values. Kvam used a related discrete shock model for common-cause failure in parallel systems [13, 3, 4, 9].

Aickin's linkset model also uses independent latent binary variables indexed by subsets. It includes one-sided tests of absent links in a fixed contingency table, including a link on the entire group. The present design ties the tables at different inspection times together through the common continuous-time rates [1].

Common-time bivariate current-status data have been studied for independence testing and copula assessment. Fisher-based monitoring-time design has also been studied for estimating a copula association parameter [6, 8, 11].

Minimal path sets and the exact configuration sum give the leading small-time completion probability in classical reliability theory. Lin, Martin and Yang use a local Hellinger-information criterion for nonregular estimation and design. Deldossi, Osmetti and Tommasi design experiments to discriminate binary copula models, while Baghel and Mondal optimize inspection schedules for cause-by-interval counts under Marshall–Olkin competing risks. Nagaraja and He compute Fisher information for order statistics and Type-II censored samples from a bivariate Marshall–Olkin model [12, 10, 5, 2, 14].

We optimize the full-snapshot squared Hellinger distance for two specified rates over one inspection time per replicate. The main proof controls the nonfull cells and localizes all first-order optimal times as the rates vanish, including pairs whose difference is arbitrarily small relative to their sum. The classical reliability coefficient then determines a closed-form limiting optimizer when $r \geq 3$. At $r = 2$, we join the regular positive-time Fisher approximation to the moving small-time boundary and exhibit a strict interior gain. The growing-dimensional comparison shows how redistributing blocks among the overlap regions can change the separation exponent at fixed global and background rates and matched marginal calibration.

2 Model and design criterion

Let $S = \{1, \dots, s\}$, where $s \geq 2$. Each block consists of $k \geq 1$ points. Each point is completed at its first independent rate-one exponential event. A block is also completed in its entirety by any active shock whose support contains it. Background supports J are proper subsets with $2 \leq |J| \leq s-1$, and their independent Poisson clocks have rates $\mu_J \geq 0$. The global shock has support S and rate λ. Block i has a dedicated shock of rate

$$\nu_i(\lambda) = a - \lambda - \sum_{J \ni i} \mu_J.$$

Assume fixed bounds $0 < a_- \leq a \leq a_+ < \infty$, $\sum_{J \ni i} \mu_J \leq a_-/4$ for every i, and $0 \leq \lambda \leq a_-/4$. In particular, $\nu_i(\lambda) \geq a_-/2$. All clocks are independent. The block size k is common to the blocks. Write

$$h_k(T) = 1 - (1 - e^{-T})^k, \qquad v_i(\lambda, T) = e^{-\nu_i(\lambda)T} h_k(T).$$

The dedicated shock and point clocks together form an independent singleton label with completion probability $1 - v_i$. Each background label is active with probability $1 - e^{-\mu_J T}$. The union of the active labels, including the global label, is the random set X_T. Each block has missing probability $e^{-aT} h_k(T)$ at every λ.

We compare $0 \leq \lambda_0 < \lambda_1 \leq a_-/4$ with the same known a, k, μ. Each replicate starts empty, and X_T is observed at a chosen deterministic $T > 0$. Let Q_λ^T be the law of X_T, $F_\lambda(T) = Q_\lambda^T(S)$, and set

$$\Delta = \lambda_1 - \lambda_0, \qquad \Lambda = \lambda_0 + \lambda_1, \qquad \zeta = \Delta/\Lambda,$$
$$\mathcal{H}_* = \sup_{T>0} H^2(Q_{\lambda_0}^T, Q_{\lambda_1}^T), \qquad H^2(P, Q) = \sum_x (\sqrt{P(x)} - \sqrt{Q(x)})^2.$$

A sequence of times is first-order optimal if its squared distance divided by $\mathcal{H}_*$ tends to one. When translating separation into testing cost, we count replicate inspections and leave exposure duration uncharged.

Let $\mathcal{E}$ contain the s singleton labels and the background labels of positive rate. Let r be the

minimum number of labels in $\mathcal{E}$ whose union is S, and define

$$w_{\{i\}} = \nu_i(0) + \mathbf{1}_{\{k=1\}}, \qquad w_J = \mu_J, \qquad A = \sum_{\substack{\mathcal{C}\subseteq\mathcal{E},\ |\mathcal{C}|=r \\ \bigcup_{J\in\mathcal{C}} J = S}} \prod_{J\in\mathcal{C}} w_J. \tag{1}$$

The label family in (1) excludes the global shock. The lower bound on A in the next theorem is a restriction on sequences of background parameters.

2.1 A common snapshot estimate

Write $H_T^2 = H^2(Q_{\lambda_0}^T, Q_{\lambda_1}^T)$, $p = F_{\lambda_0}(T)$, and $D = F_{\lambda_1}(T) - p$. The full cell is the snapshot $X_T = S$. Denote the contribution to H_T^2 from the remaining cells by R_{nf}, and their contribution to Fisher information at λ by $J_{\mathrm{nf},\lambda}$.

Lemma 1 (Full and nonfull cells)**.** *For every allowed $s, k, \lambda_0, \lambda_1$ and $T > 0$,*

$$\partial_\lambda F_\lambda(T) = T \Pr_\lambda(|S \setminus X_T| \geq 2), \qquad 0 \leq D \leq \Delta T \min(1, s e^{-a_- T}). \tag{2}$$

If

$$L_s(T) = 2sT^2 \left(e^{-a_- T/2} + \frac{1}{e^{a_- T/2} - 1} \right), \tag{3}$$

then $J_{\mathrm{nf},\lambda}(T) \leq L_s(T)$ and

$$H_T^2 = \frac{D^2}{(\sqrt{p+D} + \sqrt{p})^2} + R_{\mathrm{nf}}(T), \qquad R_{\mathrm{nf}}(T) \leq \frac{\Delta^2}{4} L_s(T). \tag{4}$$

In particular, $L_s(T)/s$ is $O(T)$ at zero, bounded over positive times, and vanishes at infinity, with bounds depending only on a_-.

Proof. For a Poisson shock of rate θ and support J, condition on all the other labels, including local completion. Its activation probability has derivative $Te^{-\theta T}$, so its contribution to the derivative of $\mathbb{E}f(X_T)$ is $T\mathbb{E}[f(X_T \cup J) - f(X_T)]$. The bracket vanishes whenever this shock has already activated. For the full-cell indicator, raising the global rate contributes $T \Pr(X_T \neq S)$. Lowering all dedicated rates subtracts $T \sum_i \Pr(X_T = S \setminus \{i\})$. This proves the identity, with right differentiation at zero. The missing-block union bound gives the inequality for D.

For a nonfull snapshot the global label is inactive. Record the other labels on the event $\mathcal{D}$ that their union is proper, weighting their product law by the global inactivity probability $e^{-\lambda T}$. The resulting subprobability projects onto the nonfull cells. Its score is $-T + S_{\mathrm{int}}$, where the singleton score has mean zero and variance $\sum_i T^2 v_i/(1 - v_i)$ under the unweighted product law, since $\partial_\lambda v_i = T v_i$. Also $\Pr(\mathcal{D}) \leq \sum_i v_i$. By $(x + y)^2 \leq 2x^2 + 2y^2$ and $v_i \leq e^{-a_- T/2}$, the restricted squared-score integral is bounded by $L_s(T)$. Grouping weights w_j into a cell can only decrease it, by $(\sum_j w_j')^2/(\sum_j w_j) \leq \sum_j (w_j')^2/w_j$, omitting identically zero weights. This proves the Fisher bound. Integrating the square-root derivatives over the parameter segment and applying Cauchy–Schwarz gives the bound on R_{nf}. The identity for H_T^2 then follows by separating the full cell. □

3 Sharp fixed-dimensional separation

Two estimates will also be used at cover order two. For fixed s, the probability of any completed block is $O(T)$, uniformly in k and the allowed rates, because each marginal completion probability is at most $(a_+ + 1)T$. Thus (2) gives

$$D = \Delta T[1 + O(T)]. \tag{5}$$

Let $B_\lambda(T)$ be the full completion probability with the global label removed while keeping the other rates at their values at λ. A background label has activation probability $\mu_J T + O(T^2)$. A singleton label has probability $(w_{\{i\}} - \lambda)T + O(T^2)$, uniformly in k, since $(1-e^{-T})^k \le T^2$ for $k \ge 2$. Partition by the exact active set. Successful configurations with exactly r labels give the coefficient of T^r; those with more labels contribute $O(T^{r+1})$ in total. Changing the singleton weights from their values at zero changes the coefficient by $O(\lambda)$. Consequently, for $A \ge A_* > 0$ and small T, λ,

$$B_\lambda(T) = AT^r[1 + O(T + \lambda)], \qquad F_\lambda(T) = 1 - e^{-\lambda T} + e^{-\lambda T} B_\lambda(T) = (\lambda T + AT^r)[1 + O(T + \lambda)]. \tag{6}$$

All constants here may depend on s, a_-, a_+, A_*.

For $T \ge \eta > 0$, independent activation of all singleton labels gives $F_\lambda(T) \ge b_\eta = (1 - e^{-a_- \eta/2})^s$. Hence (2) and (4) give

$$H_T^2/\Delta^2 \le s^2 T^2 e^{-2a_- T}/(4b_\eta) + L_s(T)/4.$$

For fixed s, η this envelope is bounded and tends to zero at infinity. The same bound without the factors $1/4$ applies to the baseline Fisher information, by (2) and the bound on $J_{\mathrm{nf},0}$.

Theorem 1 (Minimum cover order)**.** *Fix s, $r \in \{3, \dots, s\}$, and $A_* > 0$. Under the model in Section 2, suppose the minimum cover order is r at every parameter point and $A \ge A_*$. As $\Lambda \downarrow 0$, allow $\zeta \in (0, 1]$ and all admissible a, k, μ to vary. Set $\alpha = 1/(r-1)$ and*

$$g_{r,A,\zeta}(t) = \frac{t}{\left(\sqrt{At^{r-1} + (1+\zeta)/2} + \sqrt{At^{r-1} + (1-\zeta)/2}\right)^2}. \tag{7}$$

The normalized distance $H_T^2/(\Delta^2 \Lambda^{\alpha-1})$ differs from $g_{r,A,\zeta}(T/\Lambda^\alpha)$ by at most $O(\Lambda^\alpha)$, uniformly over all $T > 0$. Define

$$\begin{aligned} q_r(\zeta) &= \frac{1 + \sqrt{1 + r(r-2)(1-\zeta^2)}}{2r(r-2)}, \\ \tau_{r,A}(\zeta) = (q_r(\zeta)/A)^\alpha, &\qquad K_{r,A}(\zeta) = \frac{\tau_{r,A}(\zeta)}{1 + 2rq_r(\zeta)}. \end{aligned} \tag{8}$$

Then $\mathcal{H}_* \sim K_{r,A}(\zeta)\Delta^2 \Lambda^{\alpha-1}$, uniformly. A sequence $T > 0$ is first-order optimal if and only if $T/(\tau_{r,A}(\zeta)\Lambda^\alpha) \to 1$. The error constants may depend on s, a_-, a_+, A_*.

Proof. Write $h = H_T^2/(\Delta^2 \Lambda^{\alpha-1})$, $t = T/\Lambda^\alpha$, and $g = g_{r,A,\zeta}$. By (4)–(6), for $T \le \eta$ with $\eta > 0$ sufficiently small,

$$|h - g(t)| \le C\big[(T + \Lambda)g(t) + \Lambda^{1-\alpha} T\big].$$

Here the full-cell numerator uses the relative estimate (5), which remains valid as $\Delta/\Lambda \to 0$. The bound $g(t) \le Ct/(1 + t^{r-1})$ bounds both $g(t)$ and $tg(t)$ uniformly for $r \ge 3$. Thus the right side is $O(\Lambda^\alpha)$, since $\alpha \le 1/2$. On $T \ge \eta$, the fixed-time envelope above gives $h = O(\Lambda^{1-\alpha})$ and $g(t) = O(t^{2-r}) = O(\Lambda^{1-\alpha})$. Combining the two regions yields

$$\sup_{T>0} \left| \frac{H_T^2}{\Delta^2 \Lambda^{\alpha-1}} - g_{r,A,\zeta}(T/\Lambda^\alpha) \right| = O(\Lambda^\alpha). \tag{9}$$

Put $q = At^{r-1}$, $b = (1-\zeta^2)/4$. The logarithmic derivative of g with respect to t on the logarithmic scale is $1-(r-1)q/\sqrt{q^2+q+b}$. The fraction increases strictly from zero to one. Its unique crossing gives $r(r-2)q^2-q-b=0$; the positive root gives the maximizing time and value in (8). At fixed s, A is bounded above by a constant A^*. On the compact parameter set $[A_*, A^*]\times[0,1]$, the peak value is bounded away from zero. Uniform decay of g at both ends and uniqueness of its maximum give a uniform positive gap outside every fixed relative neighborhood of the maximizing time. The uniform bound (9) transfers the maximum and this optimality criterion to the original experiment. □

Examples. Without background shocks $r=s$ and $A=(a+\mathbf{1}_{\{k=1\}})^s$. For a nonempty core C and $d\geq 3$ disjoint nonempty petals P_i partitioning S with C, suppose the only background supports are $C\cup P_i$ and the rates β_i have a fixed positive lower bound. Then $r=d$, because each petal requires a separate label. A cover using d labels takes either the background label or, for a singleton petal, its internal singleton label. At least one background label is required for the core. Thus $A=\prod_i(\beta_i+u_i)-\prod_i u_i$, where $u_i=\mathbf{1}_{\{|P_i|=1\}}(a-\beta_i+\mathbf{1}_{\{k=1\}})$.

4 The second-order boundary

Theorem 2 (Second-order boundary)**.** *In the model of Section 2, fix $s\geq 2$ and $A_*>0$. Suppose the minimum nonglobal cover order equals two at every parameter point and its coefficient in (1) satisfies $A\geq A_*$. As $\Lambda\downarrow 0$, all allowed a,k,μ and $\zeta\in(0,1]$ may vary. For $T>0$, write*

$$J_0(T)=\sum_{x\subseteq S}\frac{(\partial_\lambda Q_\lambda^T(x)|_{\lambda=0})^2}{Q_0^T(x)},\qquad J_0(0)=1/A,\qquad J_0(\infty)=0,\qquad J_*=\max_{T\in[0,\infty]}J_0(T).$$

Then $\mathcal{H}_*/\Delta^2=J_*/4+o(1)$ uniformly as $\Lambda\downarrow 0$. A sequence $T>0$ is first-order optimal if and only if

$$\Lambda/T\to 0\qquad \textit{and}\qquad J_*-J_0(T)\to 0. \tag{10}$$

Here J_0 and J_* use the current background and singleton parameters at each point of the sequence.

Proof. At each positive time, every cell can occur through an exact singleton configuration with all nonsingleton labels inactive. On each compact interval of positive times this gives a uniform positive lower bound at fixed s: the total background rate is at most $sa_-/8$ and $e^{-(a_++1)T}\leq v_i\leq e^{-a_-T/2}$. The finite label probabilities have uniformly bounded first and second parameter derivatives there. Integrating the square-root derivatives thus gives $H_T^2/\Delta^2=J_0(T)/4+O(\Lambda)$ uniformly on the interval, including arbitrarily small Δ/Λ. At zero, (2), (6), and $J_{\mathrm{nf},0}(T)\leq L_s(T)$ give $J_0(T)=1/A+O(T)$. The envelope above Theorem 1 gives uniform decay at infinity for both normalized separation and Fisher information. This justifies the compactification and gives uniform positive lower and finite upper bounds for J_*.

For all $T>0$, define the nonnegative loss

$$P(T)=\frac{1}{4A}-\frac{1}{(\sqrt{A+\lambda_0/T}+\sqrt{A+\lambda_1/T})^2}.$$

In fact,

$$\frac{H_T^2}{\Delta^2}=\frac{J_0(T)}{4}-P(T)+o(1)\quad\text{uniformly for all } T>0. \tag{11}$$

On $T \le \eta$, (4)–(6) with $r = 2$ give $H_T^2/\Delta^2 = 1/(4A) - P(T) + O(\eta + \Lambda)$, while $J_0(T) = 1/A + O(\eta)$. On $T \ge \eta$, use the compact-interval expansion and the common vanishing tails, together with $P(T) = O(\Lambda/\eta)$. Taking $\Lambda \downarrow 0$ and then $\eta \downarrow 0$ proves (11).

The loss $P(T) \to 0$ along a sequence if and only if $\Lambda/T \to 0$, using $\lambda_1 \ge \Lambda/2$ and $A \in [A_*, A^*]$. A maximizer $T_* \in [0, \infty]$ of the current J_0 is finite, since $J_0(0) > J_0(\infty)$. At $T = \max(T_*, \sqrt{\Lambda})$, the loss tends to zero and clipping costs at most $O(\sqrt{\Lambda})$ in J_0 by its uniform zero-time expansion. This attains the upper bound $J_*/4 + o(1)$ in (11). Finally, the gap from $J_*/4$ is $(J_* - J_0(T))/4 + P(T) + o(1)$. The two nonnegative terms must vanish exactly under the two conditions in (10). □

For two blocks without background, let $v = e^{-aT} h_k(T)$. The zero-rate law is the product of the two margins and the four cell derivatives are $Tv^2, -Tv^2, -Tv^2, Tv^2$. Thus $J_0(T) = T^2 v^2/(1-v)^2$. With $c = a + \mathbf{1}_{\{k=1\}}$ we have $v \le e^{-cT}$ and $J_0(T) < 1/c^2$ at every positive time. Its maximum is $1/c^2$ at zero. Theorem 2 gives $\mathcal{H}_* \sim \Delta^2/(4c^2)$ and the complete optimal window $\Lambda \ll T \ll 1$. These conclusions hold uniformly over allowed a, k, since the strict gap on any compact interval of positive times follows from the bound by $T^2/(e^{cT} - 1)^2$ and compactness in c.

A strict interior gain. Take a fixed $s = 512$, $a = a_- = a_+ = 1$, $k = 2$, a core of 508 blocks and two petals of two blocks each. The only backgrounds are the unions of the core with each petal, each at rate $1/8$. The sole two-label cover uses both background labels, so $A = 1/64$. At $T = 1/2$ keep the pattern on the other blocks when a chosen core block is missing and otherwise record one symbol. The missing-core event has constant probability $\ell = e^{-1/2} h_2(1/2) > 63/125$, using $e^{-1/2} > 3/5$. Given that event, the background and global shocks are inactive, and the remaining singleton labels are independent with $v_i(0, 1/2) \ge \ell$. The Fisher information of this coarsened experiment is at least

$$511\,(63/125)\,(1/4)\,(63/62) > 64.$$

Indeed, the Fisher information of each conditional Bernoulli variable is $T^2 v_i/(1 - v_i)$. Grouping cells contracts Fisher information by Cauchy–Schwarz, so $J_0(1/2) > 64 = J_0(0)$. The limiting normalized separation exceeds $1/(4A)$, and all first-order optimal sequences are bounded away from both endpoints of the time axis.

5 An all-time bound in growing dimension

A common block in all positive-rate backgrounds permits a lower bound from snapshots in which that block remains incomplete. This gives a second source of separation in addition to the all-complete cell.

Theorem 3 (All-time comparison)**.** *Under Section 2, allow s, k and the rates to vary. Suppose some block belongs to every positive-rate background support, and assume $0 < \Delta \le \Delta_* := \min\{1/64, a_-/4, a_-^2/4096\}$. With $p(T) = F_{\lambda_0}(T)$,*

$$\mathcal{H}_* \asymp \min\left\{1, s\Delta^2 + \sup_{\{T>0:\, p(T) \le 1/16\}} \frac{\Delta^2 T^2}{\Delta T + p(T)}\right\}. \tag{12}$$

Comparison constants depend only on a_-, a_+. The anchor condition is vacuous when all background rates are zero.

Proof. By Lemma 1, the nonfull contribution is $O_{a_-}(s\Delta^2)$, uniformly in T. The full contribution is at most $D^2/(p + D)$, which is increasing in $D \ge 0$. For $p \le 1/16$, inserting $D \le \Delta T$ bounds it

by $G(T) := \Delta^2 T^2/(p + \Delta T)$. For $p > 1/16$, use $D \le \Delta T \min(1, se^{-a_- T})$ and $\min(1, x)^2 \le x$. The contribution is then at most $16 s\Delta^2 T^2 e^{-a_- T} = O_{a_-}(s\Delta^2)$. Taking the supremum and using $H_T^2 \le 2$ gives the upper bound in (12).

At $T = 1$ a chosen anchor is missing with probability $e^{-a} h_k(1) \ge e^{-a_+ - 1}$, independent of λ. On this event all nonsingleton clocks are inactive, and the other $s - 1$ singleton labels remain independent. The square root of the missing probability of each singleton has parameter derivative $\sqrt{v_i}/2 \ge e^{-(a_+ + 1)/2}/2$. Its squared Hellinger distance is therefore at least a constant times Δ^2. Multiplication of the conditional affinities and retention of the anchor event give a lower bound $c \min(1, s\Delta^2)$ in the full experiment. This settles the case $s\Delta^2 \ge 1$ by $H^2 \le 2$.

For the remaining lower bound suppose $s\Delta^2 < 1$ and $p \le 1/16$. The inequality $1 - p \le se^{-a_- T}$ gives $T \le a_-^{-1} \log(16s/15)$. Thus $\Delta T \le 3\sqrt{\Delta}/a_- \le 3/64$, using $s \le \Delta^{-2}$ and $\Delta \log(1/\Delta) \le \sqrt{\Delta}$. Equation (2) then ensures $F_{\lambda_0 + u}(T) \le 1/8$ along the path.

Under any law along that path let m be the common missing probability. Monotone events in the same direction for independent label activations are positively associated. This follows by induction on the number of labels: decompose covariance conditional on the last Bernoulli coordinate and use the common monotonicity of the conditional means [7].

If $m \ge 1/4$, two chosen blocks are jointly missing with probability at least m^2. If $m < 1/4$, add blocks in order until the probability that all chosen blocks are complete first reaches $1/2$ or less. The probability drops by at most m per step, so at this step the chosen group has completion probability in $(1/4, 1/2]$ and is proper because $F \le 1/8$. Association of completion events implies that the complementary group completes with probability at most $1/2$. Association of the two noncompletion events gives joint noncompletion probability at least $1/4$. In both cases the probability of two or more missing blocks is at least $1/16$.

Integrating (2) yields $\Delta T/16 \le D \le \Delta T$. The full cell in (4) is then at least $cG(T)$, since its square-root denominator is bounded by $4(p + \Delta T)$. Taking suprema and combining the two separate lower bounds gives (12). □

6 Two overlapping background shocks

Partition $S = C \sqcup P_1 \sqcup P_2$ into a nonempty core and two nonempty regions. The only possible backgrounds are $C \cup P_i$ at known rates $\beta_i \ge 0$ with $B = \beta_1 + \beta_2 \le a_-/4$. Let $\ell = \log s$ and $L = \max(0, \log(1/\Lambda))$. The formulas below apply to a comparison at matched $s, a, k, \lambda_0, \lambda_1, \beta_1, \beta_2$. Dedicated rates follow the balancing rule and can differ across geometries.

Theorem 4 (Overlap geometry)**.** *Retain the model of Section 2, the partition and background assumptions above, and $0 \le \lambda_0 < \lambda_1 \le a_-/4$ with $\Delta \le \Delta_*$ as in Theorem 3.*

1. *If $|P_1| = |P_2| = 1$ and $s \ge \max(4096, 16L^2)$, then*

$$\mathcal{H}_* \asymp \min\{1, s\Delta^2 + R_C\}, \qquad R_C = \frac{\Delta^2 \ell}{\Lambda + B}. \tag{14}$$

2. *If $|P_i| \ge \eta s$ for both i, where $0 < \eta < 1/2$, and $s \ge \max(4096, 4L^2/\eta^2)$, then*

$$\mathcal{H}_* \asymp \min\{1, s\Delta^2 + R_P\}, \qquad R_P = \frac{\Delta^2 \ell^2}{\Lambda \ell + D_s}, \qquad D_s = (1 \wedge \beta_1 \ell)(1 \wedge \beta_2 \ell). \tag{15}$$

Both comparisons are two-sided up to multiplicative constants depending only on a_-, a_+. The parameter η may vary subject to the stated dimension bound. In each case one may take $T = 1$ when $s\Delta^2 \ge 1$ or the displayed $R \le s\Delta^2$. Otherwise $T = \ell/[2(a_+ + 1)]$ attains the optimal order.

Proof. Apply Theorem 3 with a core block as the anchor. Write $b_i = 1 - e^{-\beta_i T}$, and let Q_C, Q_i be the probabilities of completing the respective regions by singleton labels at λ_0. On $p(T) \le 1/16$, the missing-block bound gives $T \le K\ell$ for $K = \max(1, 2/a_-)$. The bound $p(T) \ge 1 - e^{-\lambda_0 T}$ then gives $\Delta T + p(T) \ge (15/32)\Lambda T$, so $G(T) \le C\Delta^2\ell/\Lambda$.

With singleton petals, if at least one background activates, at most one singleton is needed. Thus $p(T) \ge (1 - e^{-BT})q_*(T)$, where $q_*(T) = 1 - e^{-a_- T/2} \ge 1 - e^{-BT}$. It follows on $p(T) \le 1/16$ that BT is bounded and $p(T) \ge cBT\min(T, 1)$. If $B > 0$, this gives $G(T) \le C\Delta^2\ell/B$. With large petals, $p(T) \ge b_1 b_2$. If both rates are positive, monotonicity of T/b_i and $T \le K\ell$ give $G(T) \le C\Delta^2\ell^2/D_s$. Combining each bound with the common Λ bound gives $\sup G \le CR_C$ or CR_P, respectively. The common bound suffices when $B = 0$ or $D_s = 0$, respectively.

For the lower bound, write $R = \Delta^2\ell^2/d$, where $d = (\Lambda + B)\ell$ or $d = \Lambda\ell + D_s$. The time $T = 1$ handles $s\Delta^2 \ge 1$ or $R \le s\Delta^2$. Otherwise use $T_s = \ell/[2(a_+ + 1)]$. Each baseline singleton has missing probability at least $s^{-1/2}$ at T_s. In the singleton-petal case, full completion requires a global activation, some background activation, or singleton completion of the core. Thus $p(T_s) \le (\lambda_0 + B)T_s + Q_C$, with $Q_C \le e^{-\sqrt{s}/2} \le \Lambda^2$ by the dimension bound. In the large-petal case, full completion requires a global activation, both backgrounds, or singleton completion of at least one petal. Hence $p(T_s) \le \lambda_0 T_s + D_s + Q_1 + Q_2$, since $b_i(T_s) \le 1 \wedge \beta_i\ell$, and $Q_i \le e^{-\eta\sqrt{s}} \le \Lambda^2$.

In both cases $p(T_s) + \Delta T_s \le d + 2\Lambda^2$. The inequality $R > s\Delta^2$ gives $d < \ell^2/s$ and, since $d \ge \Lambda\ell$, also $\Lambda^2 \le d/s$. Consequently $p(T_s) < (1 + 2/s)\ell^2/s < 1/16$ for $s \ge 4096$, and $G(T_s) \ge cR$. The pointwise lower bound from Theorem 3 completes the proof. □

Same rates, different separation. Let $a = 1$, use the same common k in both models, and take $\lambda_0 = 0, \lambda_1 = \delta$, $s \asymp \delta^{-1/2}$, and $\beta_1 = \beta_2 = \delta^{3/8}$. Compare a core of size $s-2$ with singleton petals against three regions of asymptotically equal size. The dimension and rate assumptions hold for sufficiently small δ. In (14) the regular term $s\delta^2 \asymp \delta^{3/2}$ dominates. In (15) each background factor is unsaturated, $D_s \asymp \delta^{3/4}\ell^2$ dominates $\delta\ell$, and $R_P \asymp \delta^{5/4}$ dominates $s\delta^2$. Hence $\mathcal{H}_{*,P}/\mathcal{H}_{*,C} \asymp \delta^{-1/4}$. Figure 2 illustrates this regime.

7 Replicate complexity and numerical illustrations

For $\varepsilon \in (0, 1/2)$, let $N_{\mathrm{fix}}(\varepsilon)$ be the smallest deterministic number of replicates, and let $N_{\mathrm{seq}}(\varepsilon)$ be the infimum of $\max_i \mathbb{E}_i N$ over procedures that stop almost surely under both candidates. Both require each error probability to be at most ε. We claim

$$N_{\mathrm{fix}}(\varepsilon) \asymp_\varepsilon N_{\mathrm{seq}}(\varepsilon) \asymp_\varepsilon 1/\mathcal{H}_*.$$

Choose a single time with $H^2 \ge \mathcal{H}_*/2$. A likelihood ratio test at threshold one has sum of errors at most the product affinity $(1 - H^2/2)^N$, giving the upper bound. The fixed-time link between Hellinger separation and replicate count is classical [15].

For the lower bound, use common stop, design and decision rules, including parameter-independent randomization in the transcript. At a continuing history h, let K_h be the kernel selecting the next time, and let $f_i(h)$ be the density of the history under candidate i against a common measure. The next replicate decreases affinity by the integral over continuing histories of

$$\sqrt{f_0(h)f_1(h)} \int \left[1 - \rho(Q^T_{\lambda_0}, Q^T_{\lambda_1})\right] K_h(dT).$$

This follows by writing the joint densities of the next time and snapshot using the same K_h under both candidates. The inner integral is at most $\mathcal{H}_*/2$. Sum over depths and use $\sqrt{f_0 f_1} \le (f_0 + f_1)/2$.

The unfinished affinity of a truncated transcript vanishes as the truncation increases by Cauchy–Schwarz and almost-sure stopping. Thus the final squared distance is at most $\mathcal{H}_*(\mathbb{E}_0 N + \mathbb{E}_1 N)/2$. The decision errors imply total variation at least $1 - 2\varepsilon$, which is bounded above by the square-root distance of the transcript laws. This proves the lower bound for N_{seq}, and $N_{\mathrm{fix}} \geq N_{\mathrm{seq}}$ supplies the other lower bound. Theorem 1 gives the exact first-order optimum for the single-snapshot Hellinger criterion. The testing result gives the replicate count up to constants for fixed error.

For the first figure we sum the full-state square-root distances directly. For the second figure we group full snapshots by the set of regions that are completely covered, then evaluate affinity by summing the product terms for the remaining region patterns.

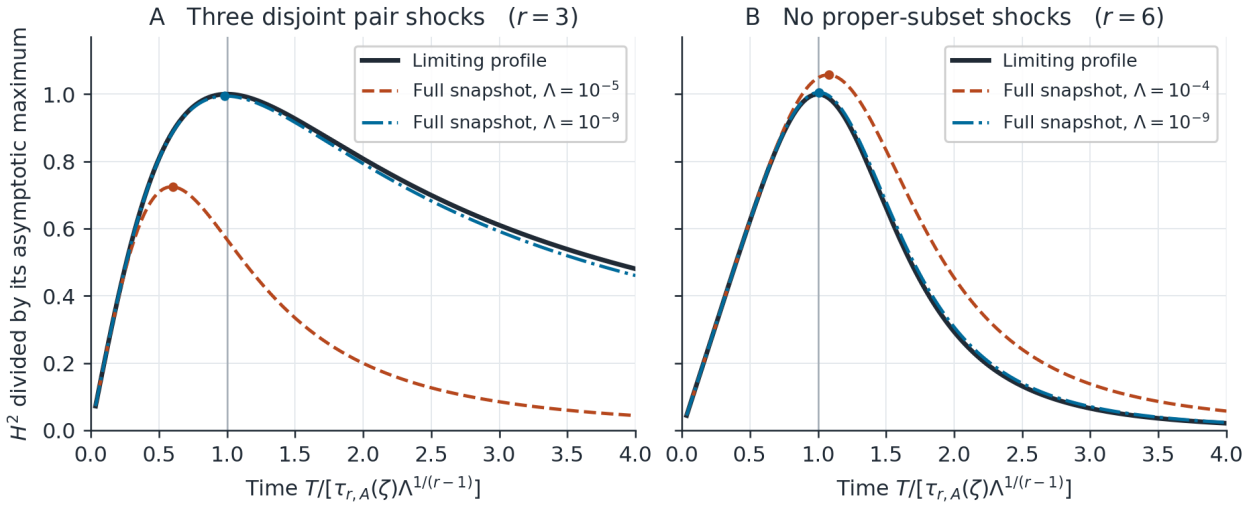


Figure 1: Fixed-dimensional profiles and numerically evaluated full-snapshot distances for three disjoint pair shocks of rate $1/5$ each ($r = 3,\ A = 1/125$) and for no background shocks ($r = 6,\ A = 64$). Both use six blocks, $a = k = 1$ and $\zeta = 0.6$. Times and distances are divided by the predicted maximizing time and height. The dots mark grid maxima.

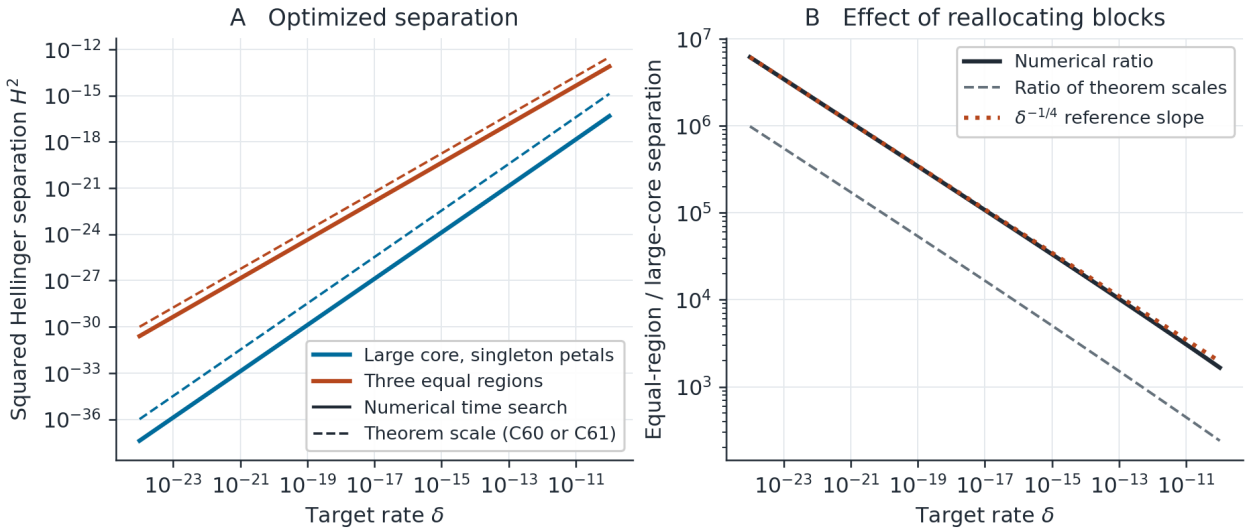


Figure 2: Two geometries with matched global and background rates and marginal calibration. Both use $s = 3\lceil \delta^{-1/2}/3 \rceil$, $a = k = 1$, $\lambda_0 = 0, \lambda_1 = \delta$, and $\beta_i = \delta^{3/8}$. The theoretical slopes are $3/2$ for the large core and $5/4$ for the large petals. The solid lines in panel A are lower bounds on the unrestricted time supremum obtained from a locally refined time grid. Dashed lines use the comparison scales in (14)–(15). Panel B takes the ratio of the searched values as a numerical illustration.

8 Discussion

At fixed dimension with $r \geq 3$, the order of the leading reliability term determines the scale on which an accidental completion and a global shock contribute comparable mass to the complete cell. Equation (4) explains why optimizing this cell alone gives the first-order optimum in that regime. At order two, finite-time differences in the remaining cells can affect the leading coefficient. The full Fisher curve then determines the optimal design.

For two overlapping backgrounds in growing dimension, a large common core can be completed by either background, leaving at most one easily completed singleton petal. With two large petals, each can remain difficult to complete unless its own background acts. This accounts for the separately saturated probabilities in (15). The two-geometry comparison requires given supports and a shared nonempty core. Questions beyond the present parameter ranges include unknown background rates, vanishing leading coefficients, additional growth regimes, and observation cost that depends on exposure time.

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